

CY 702: Advanced Inorganic Chemistry: Theory and Applications

Course Code	CY 702
Course Name	Advanced Inorganic Chemistry: Theory and Applications
L-T-P-C	3-0-0-3
Prerequisites	Consent of the faculty member
Intended for	Ph.D.
Distribution	Elective
Approval	1st Adhoc Senate (Apr 2011)

Course Contents

- **Structure and Properties of Solids**
Bonding in metals, Band theory, Density of States, k space and Brillouin Zones; Ionic, covalent and hydrogen bonded solids; electronic properties of solids, conductors, semiconductors, insulators, ferroelectricity, anti-ferroelectricity, piezoelectricity. [10 Lectures]
- **Chemistry of Transition elements and Coordination Chemistry**
Limitations of VB theory, crystal field theory, crystal field diagrams, ligand field theory, molecular orbital theory; spectrochemical series, nephelauxetic series; structural distortion and lowering of symmetry, electronic, Jahn-Teller effects on energy levels. [10 Lectures]
- **Spectral and Magnetic Properties of Complexes**
Spectroscopic ground states; Orgel energy level and Tanabe-Sugano diagrams for transition metal complexes; Charge transfer spectra; electronic spectra of octahedral and tetrahedral complexes and calculation of ligand-field parameters. Types of magnetic behaviors- magnetic susceptibility measurements- Gouy method, diamagnetic corrections- spin only value- orbital contribution- spin orbit coupling- ferro and antiferromagnetic coupling- Application of magnetic measurements to structure determinations of transition metal complexes. [8 Lectures]
- **Instrumental Methods for Studying Inorganic Compounds**
Optical activity in coordination complexes- ORD and CD- cotton effect- applications. Infrared spectroscopy: Nuclear Magnetic Resonance- Mossbauer and Electron Spin Resonance Spectroscopy for structural studies of complexes- Importance of molar conductance studies in coordination chemistry. [6 Lectures]
- **Synthetic Methodologies and Supramolecular Chemistry**
Ligand design and ligand synthesis: polypyridine, Schiff base, oxime, macrocycle etc. ligands, molecular forces, self-assembly, host / guest chemistry, molecular recognition molecular receptors, supramolecular devices, supramolecular frameworks, molecular machines, molecular/crystal engineering. [6 Lectures]

Text Books

1. A.R. West, **Solid State Chemistry and its Applications**, Wiley India Pvt Ltd (2007)
2. L.V. Azaroff, **Introduction to Solids**, Tata McGraw Hill (1984)
3. F. A. Cotton, G. Wilkinson, C. M. Murillo and M. Bochmann, **Advanced Inorganic Chemistry**, 6th Edition, Wiley India Pvt Ltd, 2007.
4. J. E. Huheey, E. A. Keiter and R. L. Keiter, **Inorganic Chemistry: Principles of Structure and Reactivity**, 4th Edition, Pearson, 2006.
5. J. D. Lee, **Concise Inorganic Chemistry**, Wiley India Pvt Ltd, 2008.
6. J. W. Steed and J. L. Atwood, **Supramolecular Chemistry**, 2nd Edition, John Wiley and Sons, 2009.